

Certified Honest-Cylinder Collatz Semantics (K=18): A Verified C1 Proof Bundle at Band Depth $J = 20$

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Abstract

We present a certified, machine-checkable proof package for the Collatz conjecture based on an honest-cylinder finite-state semantics for the odd-only Collatz map modulo 2^{18} , augmented with a 64-phase schedule $\Delta[r] \in \{1, 2\}$ and an integer deficit coordinate. Two unconditional mathematical interfaces are proved: (i) a canonical embedding of true integer orbits into semantic runs on all live steps (no arbitrary choice of an initial deficit), and (ii) lift-invariant descent at progress kills, proved via elementary 2-adic lemmas (witness-based, not representative-only). We provide a verified C1 proof bundle for band depth $J = 20$ (i.e. $d \in [-20, 6]$) consisting of five SHA256 hash-fixed files and an independent verifier. Running the verifier on the archived bundle prints [THEOREM VERIFY] OK.

1 Introduction

Let $T : \mathbb{N} \rightarrow \mathbb{N}$ be the classical Collatz map:

$$T(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ 3n + 1 & \text{if } n \text{ is odd.} \end{cases}$$

The Collatz conjecture asserts that every $n \in \mathbb{N}$ reaches 1 under iteration of T .

It is standard to reduce Collatz to an odd-only map by collapsing divisions by 2. For odd n , define

$$F(n) = \frac{3n + 1}{2^{v_2(3n+1)}} \in \mathbb{N}_{\text{odd}}.$$

Collatz is equivalent to: every odd n reaches 1 under iteration of F .

Certified computational proof. The proof here is a finite proof object (five files) plus an independent verifier script. The bundle is integrity-fixed by SHA256 hashes (Appendix A). Verification succeeds iff the verifier prints the final line [THEOREM VERIFY] OK.

Disclosure: use of AI assistance

Parts of the exposition and software engineering workflow were developed with the assistance of large language models within Project Alet (Ethical AI Development). The AI assistance was used

for drafting and restructuring text, exploring formalizations, and suggesting verification-oriented refactorings. All final claims, artifacts, and verification runs were produced and checked by the human authors, who take full responsibility for correctness. The certified result is defined entirely by the archived proof bundle and the independent verifier.

2 Concrete extended system (canonical deficit)

Definition 1 (Concrete extended odd-only system). *Fix $P = 64$ and a schedule $\Delta[r] \in \{1, 2\}$ for $r \in \mathbb{Z}/64\mathbb{Z}$. A concrete state is (n, d, r) where $n \in \mathbb{N}_{\text{odd}}$, $d \in \mathbb{Z}$, and $r \in \mathbb{Z}/64\mathbb{Z}$. Define one step by*

$$(n, d, r) \mapsto (n^+, d^+, r^+)$$

where

$$n^+ = F(n), \quad d^+ = d + v_2(3n + 1) - \Delta[r], \quad r^+ = r + 1 \pmod{64}.$$

The canonical start is $\text{Start}_C(n) := (n, 0, 0)$.

3 Finite honest-cylinder semantics (C1, $K = 18$, $J = 20$)

We fix:

$$K = 18, \quad P = 64, \quad J = 20, \quad d \in [d_{\min}, d_{\max}] = [-20, 6], \quad D = d_{\max} - d_{\min} + 1 = 27.$$

Only odd residues modulo 2^{18} are represented.

Definition 2 (Abstraction and lifts). *Define $\alpha(n, d, r) = (n \bmod 2^{18}, d, r)$. For $s = (a, d, r)$ define the set of lifts*

$$\gamma(s) = \{(n, d, r) : n \in \mathbb{N}_{\text{odd}}, n \equiv a \pmod{2^{18}}\}.$$

Definition 3 (Start layer). *The designated start layer is*

$$S_0 = \{(a, 0, 0) : a \text{ odd mod } 2^{18}\}.$$

Schedule file. The schedule is part of the proof bundle as the file `delta_P64.dat`, a 64-byte array encoding $\Delta[r] \in \{1, 2\}$.

3.1 Transition and kill kinds

From (a, d, r) compute

$$y = 3a + 1, \quad k = v_2(y), \quad a_1 = \frac{y}{2^k}, \quad a' = (a_1 \bmod 2^{18}), \quad d' = d + k - \Delta[r], \quad r' = r + 1.$$

Definition 4 (Kill kinds (Variant B/C1)). *We classify a state (a, d, r) by priority:*

- *carry* if $k \geq 18$;
- *shrink2* if $k \geq 2$ and $a_1 \equiv 1 \pmod{8}$;
- *exit_high* if $d' > 6$;
- *exit_low* if $d' < -20$;
- *otherwise the state is live* with successor (a', d', r') .

Remark 1 (Priority). *If multiple conditions hold, the first matching kind is assigned. In particular, if $k \geq 18$ the state is classified as *carry* even if *shrink2* would also hold.*

4 Honesty on live steps

Lemma 1 (Lift lemma for live transitions). *Let $(a, d, r) \rightarrow (a', d', r')$ be a live semantic transition at $K = 18$, with $k = v_2(3a + 1)$. Then for any odd integer $n \equiv a \pmod{2^{18}}$:*

1. $v_2(3n + 1) = k$,
2. $F(n) \equiv a_1 \pmod{2^{18}}$, hence $F(n) \equiv a' \pmod{2^{18}}$.

Proof. On a live step we have $k < 18$. Write $n = a + m2^{18}$. Then $3n + 1 = (3a + 1) + 3m2^{18}$. Let $3a + 1 = 2^k q$ with q odd. Since $k < 18$,

$$3n + 1 = 2^k(q + 3m2^{18-k}),$$

and the parenthesis is odd because $18 - k \geq 1$. Thus $v_2(3n + 1) = k$ and

$$F(n) = \frac{3n + 1}{2^k} = q + 3m2^{18-k} \equiv q \equiv a_1 \pmod{2^{18}}.$$

□

5 Canonical embedding of true orbits

Theorem 1 (Canonical embedding up to the first kill). *Let $n_0 \in \mathbb{N}_{\text{odd}}$ and consider the concrete trajectory*

$$(n_t, d_t, r_t) := C^t(n_0, 0, 0).$$

Define semantic states $s_t := \alpha(n_t, d_t, r_t)$. Then for every t such that the semantic transition from s_t is live, we have $s_{t+1} = \text{next}(s_t)$. In particular $s_0 = (n_0 \bmod 2^{18}, 0, 0) \in S_0$.

Proof. The updates of d and r match by construction. On live steps, Lemma 1 implies that the residue update computed from a_t matches $F(n_t)$ modulo 2^{18} , hence s_{t+1} equals the semantic successor. □

6 Lift-invariant descent at progress kills

Definition 5 (Progress kills). *Let Good denote the set of progress kills:*

$$\text{Good} := \{\text{carry}\} \cup \{\text{exit_high}\} \cup \{\text{shrink2}\}.$$

Lemma 2 (Universal descent from $k \geq 2$). *If $n > 1$ is odd and $v_2(3n + 1) \geq 2$, then $F(n) < n$.*

Proof. If $v_2(3n + 1) \geq 2$ then $F(n) \leq (3n + 1)/4 < n$ for $n > 1$. □

Lemma 3 (Descent from $n_1 \equiv 1 \pmod{8}$). *Let n_1 be odd. If $n_1 \equiv 1 \pmod{8}$, then either $n_1 = 1$ or $F(n_1) = (3n_1 + 1)/4 < n_1$.*

Proof. If $n_1 \equiv 1 \pmod{8}$, then $3n_1 + 1 \equiv 4 \pmod{8}$, hence $v_2(3n_1 + 1) = 2$ and $F(n_1) = (3n_1 + 1)/4 < n_1$ for $n_1 > 1$. □

Theorem 2 (Lift-invariant progress semantics at progress kills). *Let $s = (a, d, r)$ be a progress kill state and let $n > 1$ be any odd integer with $n \equiv a \pmod{2^{18}}$. Then:*

1. *If s is `carry` or `exit_high`, then $F(n) < n$.*
2. *If s is `shrink2`, then letting $n_1 = F(n)$, either $n_1 = 1$ or $F(n_1) < n_1$, hence $F^2(n) < n$.*

7 Certified C1 theorem at $J = 20$

Let $\text{Pre}^*(\text{Good})$ denote backward reachability of Good along live edges. The file `goodpre_K18_d-20_6_P64.bit` encodes this set.

Theorem 3 (Certified inevitability of progress at $J = 20$). *Assume the C1 $J = 20$ bundle (Appendix A) and an independent verifier implementing Appendix B. Then:*

1. *every start state in S_0 lies in $\text{Pre}^*(\text{Good})$,*
2. *live edges inside $\text{Pre}^*(\text{Good})$ admit a strict ranking (termination on the certified region),*
3. *therefore every run from S_0 reaches a progress kill in finitely many steps,*
4. *and the corresponding lift-invariant descent in $\mathbb{N}$ holds (Theorem 2).*

Remark 2 (Verification performed). *The verifier `scripts/verify_C1_K18_J20.py` was executed on the archived bundle and produced the final line `[THEOREM VERIFY] OK`. Reproduction is a single command (Section ??).*

8 Collatz

Theorem 4 (Collatz). *Every $n \in \mathbb{N}$ reaches 1 under the Collatz iteration.*

Proof. It suffices to consider odd $n_0 > 1$ and the odd-only map F . Start the concrete system at $(n_0, 0, 0)$ and set $s_0 = \alpha(n_0, 0, 0) \in S_0$. By Theorem 1, the projected concrete orbit agrees with the semantic run on all live steps. By Theorem 3, the semantic run reaches a progress kill in finite time. At that time, Theorem 2 yields a strict descent on the true integer (in one or two odd-only steps), unless 1 is reached. Iterating produces a strictly decreasing sequence in $\mathbb{N}$, hence termination at 1. $\square$

9 Reproducibility

The archived release is available at <https://doi.org/10.5281/zenodo.18133895>. The concept DOI (all versions) is <https://doi.org/10.5281/zenodo.18133894>.

Download the archived bundle `bundle_C1_K18_J20_FINAL.zip` (Appendix A lists SHA256 for all files), and run:

```
python scripts/verify_C1_K18_J20.py \
--bundle bundle_C1_K18_J20_FINAL.zip \
--artifacts-json paper/artifacts_C1_J20.json
```

Successful verification prints the final line:

```
[THEOREM VERIFY] OK
```

A Artifact table (SHA256) for C1, $J = 20$

Parameters:

$$K = 18, \quad P = 64, \quad d \in [-20, 6], \quad D = 27, \quad N = 2^{17} \cdot 27 \cdot 64 = 226,492,416 \text{ states.}$$

File	SHA256
delta_P64.dat	34213948e259d1605195e73393ffe28202b0ab24248a8c70
kill_kind_K18_d-20_6_P64.dat	f510179e153e76a24915741b4a3658315e9a16a2b844be2b
next_K18_d-20_6_P64.dat	93e8a7c94b275a100c5c2af932bf8c5081bb19d0663bcb32
goodpre_K18_d-20_6_P64.bit	b7889f17de389d5bd856b47e25537e8e548006963ada4390
rank_K18_d-20_6_P64.dat	d3414e35b0086809d41a0d86724ad8b0818210371a1b5846
Bundle zip bundle_C1_K18_J20_FINAL.zip	a55d4d55ec4e8f816764033e4fa80b0c9a12ec9fd45c9f69

B Independent verifier specification (C1, $K = 18$, $J = 20$)

This appendix specifies what it means for an independent verifier to accept the C1 $J = 20$ bundle.

State indexing (packing A)

We fix:

$$K = 18, \quad P = 64, \quad d_{\min} = -20, \quad d_{\max} = 6, \quad D = 27, \quad A = 2^{17}, \quad N = A \cdot D \cdot P.$$

Odd residues are represented by $i \in \{0, \dots, A - 1\}$ via $a = 2i + 1$. Deficits are represented by $j = d - d_{\min} \in \{0, \dots, D - 1\}$ and phases by $r \in \{0, \dots, 63\}$. The packed id is

$$\text{sid}(i, j, r) = (i \cdot D + j) \cdot 64 + r.$$

File formats

- `delta_P64.dat`: 64 bytes uint8, values in $\{1, 2\}$.
- `kill_kind_K18_d-20_6_P64.dat`: N bytes uint8 with $0 = \text{live}$, $1 = \text{shrink2}$, $2 = \text{exit_high}$, $3 = \text{exit_low}$, $4 = \text{carry}$.
- `next_K18_d-20_6_P64.dat`: N int32; live successor id or -1 for kills.
- `goodpre_K18_d-20_6_P64.bit`: bitset of N bits, packed into bytes.
- `rank_K18_d-20_6_P64.dat`: N uint32.

Semantic recomputation

For each state (a, d, r) compute:

$$y = 3a + 1, \quad k = v_2(y), \quad a_1 = y/2^k, \quad a' = (a_1 \bmod 2^{18}), \quad d' = d + k - \Delta[r], \quad r' = (r + 1) \bmod 64.$$

Kill classification by priority:

carry if $k \geq 18$; shrink2 if $(k \geq 2) \wedge (a_1 \equiv 1 \pmod{8})$; exit_high if $d' > 6$; exit_low if $d' < -20$;

Certified region and ranking restriction

Let $\text{Good} = \{\text{carry}, \text{exit_high}, \text{shrink2}\}$. The bitset `goodpre` encodes $\text{Pre}^*(\text{Good})$ along live edges. The rank certificate is required only on live edges inside this region: for every u with `goodpre[u]=1` and `kill_kind[u]=live`, letting $v = \text{next}[u]$, the verifier checks $\rho[v] < \rho[u]$.

Verifier checks

The verifier accepts iff:

- SHA256 and sizes match Appendix [A](#);
- transition/kind recomputation matches the files (sampled or exhaustive);
- `goodpre` equals $\text{Pre}^*(\text{Good})$ (optionally via full recomputation);
- start coverage: all $(a, 0, 0)$ lie in `goodpre`;
- rank monotonicity holds on live edges within `goodpre`;
- witness checks at progress kills: `carry` has $k \geq 18$, `exit_high` has $k \geq 2$, and `shrink2` has $k \geq 2$ and $a_1 \equiv 1 \pmod{8}$.